

K HARSHAVARDHAN



+91 8019252777



hvckari@gmail.com



Gangaram , Hyderabad , Telangana



<https://www.linkedin.com/in/harsha-varadhan-146181217/>

Enthusiastic B.Tech Cyber Security student at VNR VJIEET with a passion for scripting and developing cybersecurity solutions. Skilled in programming languages and cybersecurity tools, with the ability to quickly apply theoretical knowledge. Seeking opportunities to leverage technical skills in a dynamic environment.

EDUCATION

2022 - 2026

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY, Hyderabad, Telangana

Bachelor of Technology in Computer Science and Engineering - Cyber Security
CGPA : 7.52/10

2020 - 2022

SRI CHAITANYA JUNIOR COLLEGE, Hyderabad, Telangana

Higher Secondary School(XII), **91.6%**

2019 - 2020

AROHAN THE COMPLETE SCHOOL , Hyderabad, Telangana

Secondary School(X), **10/10**

PUBLICATIONS

IEEE Conference Paper – Moving Target Defense Framework for Resilient DDoS Mitigation using Dynamic Proxy Shuffling and ChaCha20 Encryption

- Designed a secure multi-server architecture integrating Moving Target Defense (MTD) and ChaCha20 encryption to mitigate large-scale DDoS attacks.
- Implemented dynamic proxy shuffling and encrypted communication to enhance service availability and attack resistance.
- Technologies: Network Security, ChaCha20 Cryptography, Proxy Servers, Distributed Systems.

Conference Paper – AMLDA 2025 – AI-Based Phishing Website Detection using Machine Learning

- Developed a real-time phishing detection system using machine learning for malicious URL classification.
- Implemented a Flask-based detection platform using TF-IDF vectorization and ML models for automated web security monitoring.
- Technologies: Python, Flask, Machine Learning, TF-IDF, Cyber Threat Detection.

PATENTS

Inventor, An Intelligent Automated Waste Segregation System with IoT-Enabled Data Monitoring and Classification Mechanism

- Indian Patent Application No: 202541117014
- Published in Patent Journal: 51/2025 (Dec 2025)
- Developed an IoT-based automated waste segregation system using sensor-based classification and real-time monitoring for smart waste management.

Inventor, A Novel Approach to Implementing ChaCha20 Encryption for Mitigating Distributed Denial of Service (DDoS) Attacks in Defense Network Architectures

- Indian Patent Application No: 202641017466
- Filed: Feb 2026
- Proposed a secure cybersecurity architecture integrating ChaCha20 encryption, proxy-based detection, and Moving Target Defense (MTD) to mitigate large-scale DDoS attacks..

PROJECTS

MEMORY FORENSICS : INVESTIGATING VOLATILE MEMORY FOR CYBER THREATS

- We had analyzed volatile memory to detect and investigate cyber threats, focusing on identifying malicious activities, such as malware or unauthorized access. This involved capturing memory dumps and using forensic tools to uncover evidence of attacks.

WASTE SEGREGATION USING ARDUINO

- This project presents a Smart Dustbin system powered by Arduino that uses IR, inductive, and moisture sensors to automatically segregate waste into metallic, wet, and dry categories, promoting efficient and hygienic disposal.

MOTAG – Intelligent DDoS Detection and Mitigation System

- Built a real-time system to monitor network traffic, detect abnormal spikes, and automatically block malicious sources to prevent DDoS attacks and maintain service availability.

AUTOMATED VULNERABILITY SCANNING TOOL

- In this project, we developed a web scanning tool that automatically analyzes a website as soon as it is opened and checks for potential issues such as vulnerabilities, errors, or security risks. The tool provides detailed reports on detected problems, helping users or developers to identify and fix them quickly. This ensures improved website security, performance, and reliability with minimal manual effort.

Enhanced Moving Target Defense Framework with ChaCha20 Encryption for Resilient DDoS Mitigation

- Built a research-driven MOTAG framework for DDoS mitigation using dynamic proxy shuffling and secure authentication. Integrated ChaCha20 and AES encryption for fast, low-overhead data protection. Designed a multi-server architecture with attack detection and filtering modules to maintain service availability. Evaluated performance through encryption latency and mitigation analysis.

SKILLS

- **Languages** : Python , C , SQL , Java , HTML
- **Tools** : Wireshark , Nmap , Security Information and Event Management (SIEM) tools , Burp Suite
- **Others** : Linux , Data Structures , Intrusion Detection Systems (IDS) , Vulnerability assessment , PEP 8 style guide , Network Security , Bash , Packet Analyzer , DDoS Mitigation & Detection
- **Frameworks** : OWASP TOP 10 , IDS/IPS
- **Platforms** : Linux , Remnux, Ubuntu, Log Analysis

INTERNSHIPS

Campus Ambassador|Learn Flu Online|Jun 2024-Jun 2024

- Organized private and public events on campus for the college community..

Cyber Security and Ethical Hacking-Microsoft by Threat-Prism | Dec 2022 - Dec 2025

- Done projects on Exploiting Server Vulnerabilities .

Cyber Security Analyst | SSEV SOFTSOLS

- Worked with securing many embedded devices and secured many websites.
-

TRAININGS & CERTIFICATIONS

- Professional Cyber Security Course | Google
 - Cyber suraksha | TATA
 - Microsoft certificate for industrial training in cyber Security
 - Bronze Certificate as a 'Smart Coder' by Smart Interviews
 - Web Development by VNR VJiet
 - "Cyber Security and Ethical Hacking" program, offered by ArIES IIT Roorkee in collaboration with Coincent
 - "A Practical Introduction to Cloud Computing" with EC-Council
 - "Introduction to Cybersecurity" course with Cisco Networking Academy
 - CEH with EC-Council
-

ROLES AND ACTIVITIES

President – VJ Garuda Vigilance (Cybersecurity Club), VNR VJiet

- Leading the cybersecurity club by organizing technical workshops, awareness sessions, and cybersecurity competitions to promote knowledge in ethical hacking and cyber defense.
- Guiding and mentoring junior members in cybersecurity fundamentals, tools, and career opportunities in the security domain.

Organizing Team Member – TechnoVista (Technical Fest), VNR VJiet

- Contributed to the planning and smooth execution of technical events and competitions as part of the college's annual technical fest.
- Coordinated with organizing teams to manage event logistics and participant engagement.

Organizing Team Member – ICAMADA (Technical Fest), VNR VJiet

- Assisted in organizing technical competitions and innovation-focused events.
- Supported event coordination, registrations, and overall event management activities.

Content Head – Dramatrix Club, VNR VJiet

- Responsible for developing creative content including scripts for skits and stage performances during college cultural events.
- Led the content team to design impactful storytelling and engaging performances.

Placement Coordinator (2025–2026), VNR VJiet

- Assisting the placement cell in coordinating recruitment drives, pre-placement talks, and interview processes.
 - Acting as a communication bridge between students, the placement cell, and recruiting companies to ensure smooth placement activities.
-

Strengths

- Strong Problem-Solving Skills
 - Quick Learner
 - Adaptability
 - Leadership and Team Management
 - Team Collaboration
-

Hobbies

- Creative Content Writing
 - Event Organizing
 - Writing and Directing Skits
-